BIOCHEMISTRY CUMULATIVE EXAM

April 4, 1998

The following technique has been developed to measure the pH gradient $\Delta pH = pH_{in} - pH_{out}$ and voltage gradient $\Delta V = V_{in} - V_{out}$ across the inner membrane of actively respiring mitochondria. Trace amounts of $^{86}Rb^+$, valinomycin, and [^{3}H] acetate were added to active mitochondria, and the radioactive ions were allowed to partition between the inside and outside of the mitochondria. The mitochondria were then filtered, and the radioactivities in the mitochondrial and filtrate fractions were measured to determine that $[^{86}Rb^+]_{in} = 10^{-9}$ M, $[^{86}Rb^+]_{out} = 10^{-11}$ M; $[acetate]_{in} = 10^{-10}$ M, $[acetate]_{out} = 10^{-12}$ M. Valinomycin is added to allow free movement of the monovalent cation $^{86}Rb^+$ across the mitochondrial membrane. The concentrations of $^{86}Rb^+$, valinomycin, and acetate are too small to affect the pH gradient or voltage across the mitochondrial membrane.

Questions.

1. (35) Calculate the voltage gradient ΔV across the inner membrane of the active mitochondria using the $^{86}Rb^+$ partition data given above. Derive any equations that are needed, and explain the calculation in detail.

2. (35) Calculate the pH gradient across the inner membrane of the active mitochondria using the [^{3}H] acetate partition data given above. You may assume that acetic acid (the protonated form of acetate) passes freely across the mitochondrial membrane and is partitioned with equal concentrations on both sides of the membrane. Derive any equations that are needed, and explain the calculations in detail.

3. (30) Calculate the free energy ΔG for transporting one mole of protons from outside to inside the mitochondrial membrane under the above conditions.

Possibly useful equations:

$\Delta G(X_{out} \rightarrow X_{in}) = RT\ln([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out})$. (Z is charge on X)
$\Delta G = \Delta G^o + RT\ln([products]/[reactants])$
$R = 2$ cal/degree.mol, $F = 23$ kcal/volt.mol
$pH = pK_a + \log([A^-]/[HA])$, pK_a (acetate) $= 4.7$